MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION

Sidhant College of Engineering

A/P Sudumbre , Tal-Maval, Dist-Pune

Microproject

Academic Year 2025-26

Program Name:- Artificial Intelligence And Machine Learning **Scheme:-** K

Subject Code:-Data Communication And Computer Network (314318) **Semester:- 4th**

A microproject on

Cyber Security

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CERTIFICATE

This is to certify that **Miss. Sakshi Diwane,** Fourth Semester of Diploma in Artificial Intelligence And Machine Learning (AIML) Engineering of Institute, Siddhant college of Engineering (Code:1624) has completed the Self Learning Activity (SLA) satisfactorily in Course Data Communication And Computer Network (314318) for the academic year 2025-2026 as prescribed in the curriculum.

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1.Introduction to Cyber Security :

Cyber security is the practice of protecting computers, networks, systems, and data from digital attacks, unauthorized access, damage, or theft. In today's modern world, almost everything is connected to the internet, such as mobile phones, laptops, banking systems, schools, businesses, and even government organizations. Because of this heavy use of technology, the risk of cyber attacks has increased rapidly.

Cyber attacks are activities carried out by hackers or cyber criminals to steal, misuse, or destroy information. These attacks can come in many forms such as viruses, malware, ransomware, phishing emails, and hacking. For example, a hacker may try to steal bank details, passwords, or personal photos. This can cause financial loss, privacy issues, and even damage to a company's reputation.

To prevent such problems, cyber security is used. It includes a combination of technologies, software, tools, and safety practices that help in protecting systems and data. Firewalls, antivirus software, encryption, and strong passwords are some common cyber security measures used to stay safe online.Cyber security is based on three main principles:

Confidentiality – ensuring that sensitive information is accessed only by authorized peopleIntegrity – making sure that data is not changed or tampered with by unauthorized users.

Availability – ensuring that systems and data are available to users whenever neededCyber security is important for everyone. For individuals, it protects personal data like photos, messages, and bank details. For businesses, it helps in protecting customer data, financial records, and important documents. For governments, it ensures the safety of national data and security systems cyber security plays a very important role in today's digital world. As technology continues to grow, the need for strong cyber security also increases. By using proper safety measures and being aware of online threats, we can protect ourselves and our data from cyber crimes.

2.Meaning of Computer Network Security :

Protecting computers, networks, and data from unauthorized access, attacks, damage, or theft.

Simple Meaning :-

It is the process of keeping your network safe so that only authorized users can access and use it.

In Easy Words :-

Computer network security is like putting locks, passwords, and guards on your internet or network to stop hackers or unwanted users.

Main Goals:Confidentiality – Data should be kept secret

Integrity – Data should not be changed wrongly

Availability – Data should be available when needed

Example:Using a password on Wi-Fi

Installing antivirus softwareUsing firewalls

Computer network security is the practice of protecting network systems and data from unauthorized access, misuse, or attacks.

Computer Network Security (Long Answer – Exam Ready)Introduction

Computer Network Security refers to the process of protecting a computer network and its data from unauthorized access, misuse, attacks, damage, or theft.

In today's digital world, networks are used everywhere (banks, colleges, companies), so protecting them is very important.

Definition:-

Computer Network Security is a set of policies, practices, and technologies used to safeguard networks, devices, and data from cyber threats and unauthorized users.

Objectives of Network Security

Confidentiality – Only authorized users can access data

Integrity – Data should not be modified without permission

Availability – Data and network should be available when needed

(These three are called the CIA Triad)

Components of Network Security

Hardware Security

Devices like firewalls, routers

Protect physical network

Software Security

Antivirus, anti-malware programs

Protect from viruses and attacks

Data Security

Encryption and backup

Protects sensitive information

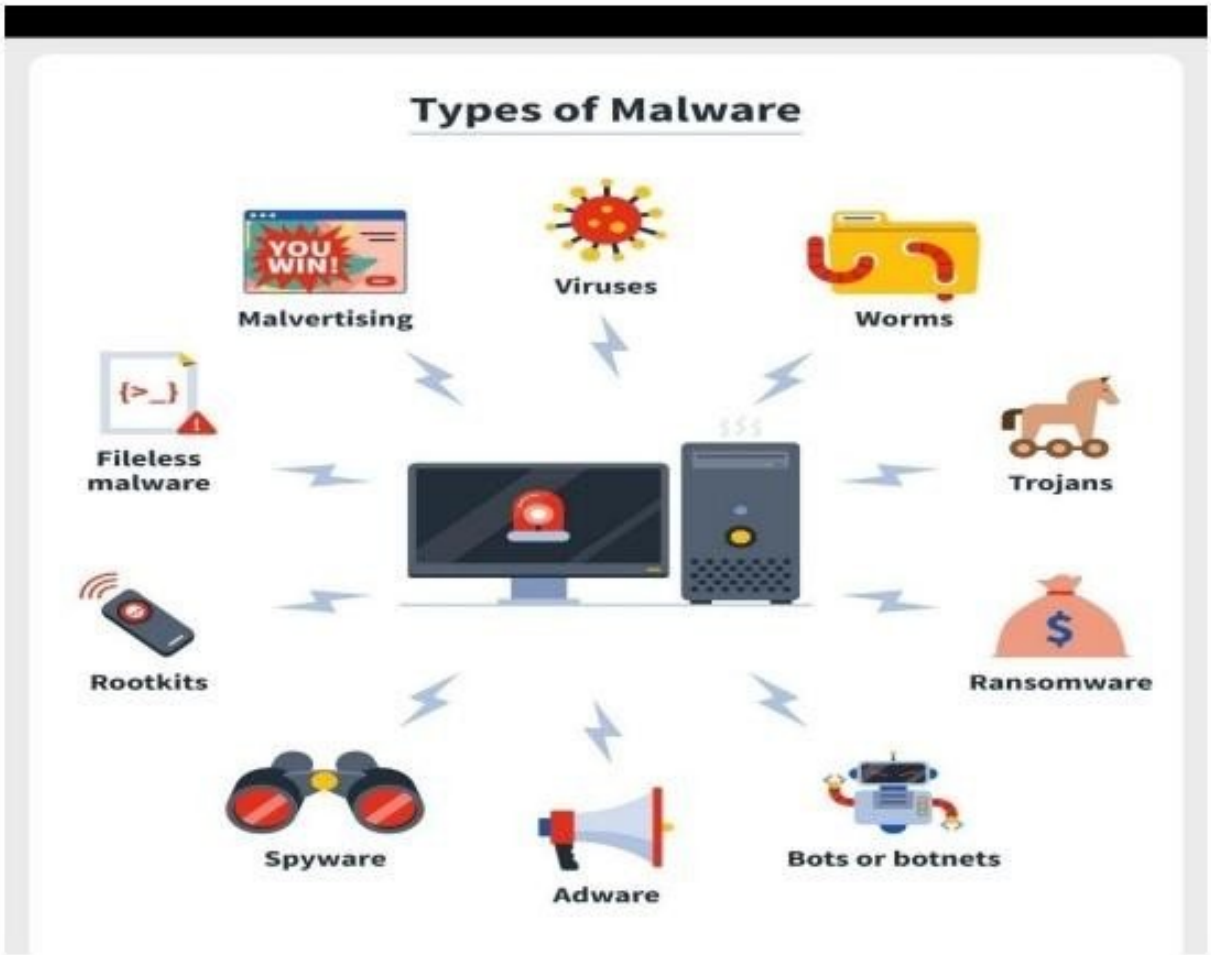
3.Types of Cyber Threats :

Cyber threats are harmful activities that try to damage, steal, or misuse data and computer systems. These threats can affect individuals, businesses, and even governments.

1. Malware (Malicious Software) :-

Malware is any harmful software designed to damage or gain unauthorized access.

Types of Malware :-



Virus – Spreads by attaching to files

Worm – Spreads automatically across networks

Trojan Horse – Disguises as safe software

2. Phishing :-

Phishing is a method where attackers send fake emails or messages to trick users into sharing personal information like passwords or bank details.

A Phishing Attack is a type of cyber attack where hackers trick users into giving sensitive information like passwords, bank details, or personal data by pretending to be a trustworthy entity.

The word “phishing” comes from “fishing,” because attackers are “fishing” for your private information.

1. How Phishing Works :-

Preparation: The attacker creates a fake website, email, or message that looks real.

Example: Fake Gmail login page, PayPal message, or bank alert.

Baiting the Victim: They send the fake email or message to the victim.

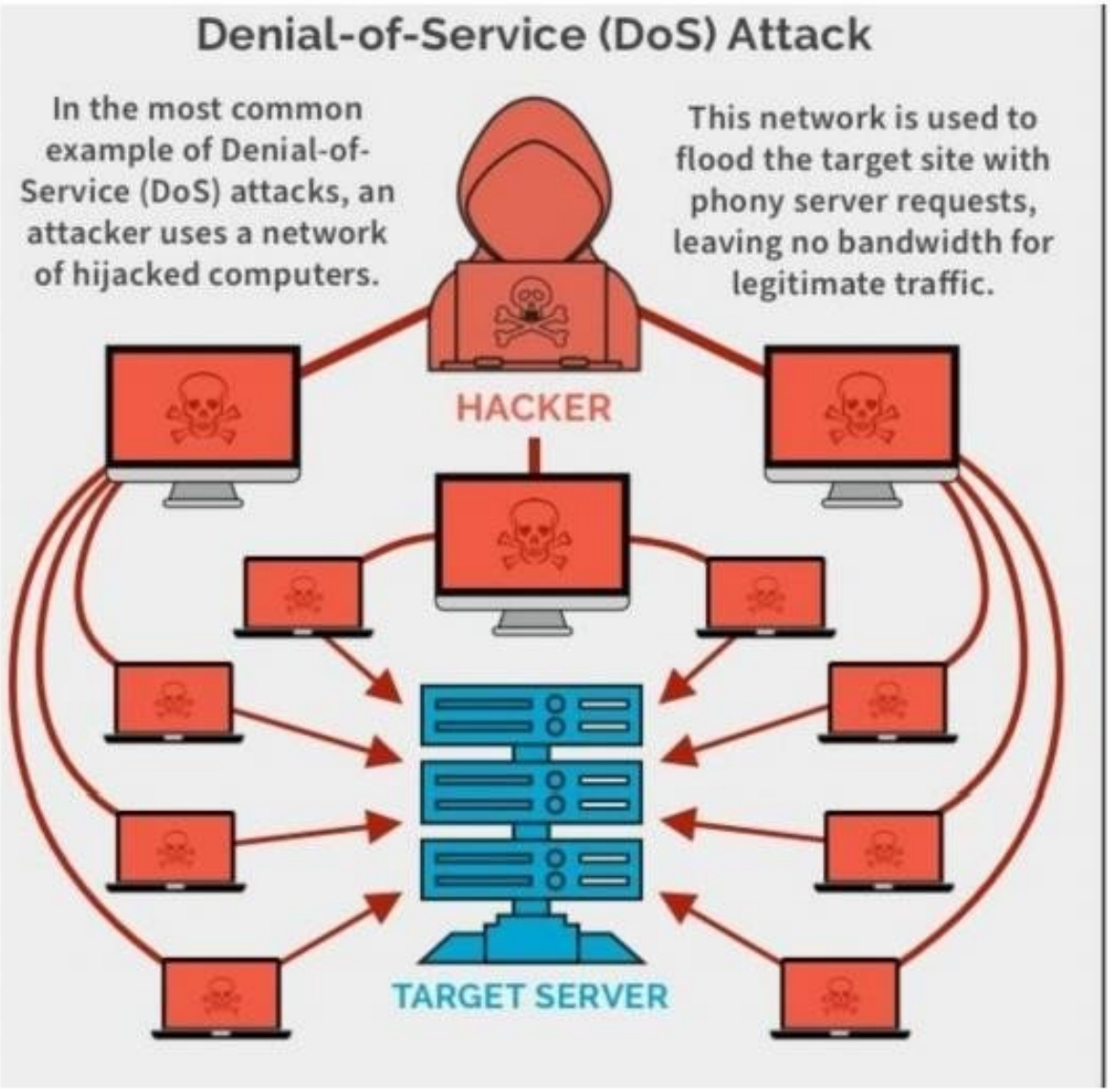
The email may have an urgent warning like “Your account will be locked!”

Capturing Information: The victim clicks the link and enters username.

3. Hacking :-

Hacking means unauthorized access to a computer or network. Hackers may steal, change, or destroy data.

4. Denial of Service (DoS) Attack :-



4. Cyber Security Techniques and Tools :-

Cyber security techniques and tools are methods and software used to protect computers, networks, and data from cyber attacks. They help in preventing unauthorized access, detecting threats, and keeping information safe.

Cyber Security Techniques :

1. Authentication :-

Authentication verifies the identity of a user.

Examples:

Passwords

OTP (One-Time Password)

Fingerprint / Face recognition

2. Encryption :-

Encryption converts data into a secret code so that only authorized users can read it.

3. Firewall Protection :-

A firewall acts like a security wall between a trusted network and untrusted networks.

4. Intrusion Detection System (IDS) :-

IDS monitors network traffic and detects suspicious activities or attacks.

5. Regular Updates & Patch Management :-

Updating software fixes security weaknesses and protects against new threats.

6. Access Control :-

Limits access to data based on user roles and permissions.

Cyber Security Tools :-

1. Antivirus Software :

Detects and removes viruses and malware.

Examples: Quick Heal, Norton

2. Firewall :

Blocks unauthorized traffic entering or leaving the network.

3. VPN (Virtual Private Network) :

Provides a secure and private connection over the internet.

4. Penetration Testing Tools :

Used by ethical hackers to test system security and find weaknesses.

5. Security Information and Event Management (SIEM) :

Collects and analyzes security data to detect threats in real time.

6. Password Managers :

Stores and manages strong passwords securely.

Conclusion :

Cyber security techniques and tools work together to protect systems and data from cyber threats. Using proper security measures like encryption, firewalls, and antivirus software helps in maintaining a safe digital environment.

Cyber security techniques include authentication, encryption, firewall protection, IDS, updates, and access control. Tools include antivirus, firewall, VPN, penetration testing tools, SIEM, and password managers.

5. Importance of Cyber Security :

Cyber security is very important in today's digital world because almost everything—banking, education, communication, shopping, and business—depends on computers and the internet. Without proper cyber security, data and systems are always at risk of attacks, theft, and damage. It helps individuals, organizations, and governments stay safe in the online environment.

1. Protection of Sensitive Data :-

Cyber security protects important information such as:

Personal details (name, address, passwords)

Bank and financial data

Business secrets

Without security, this data can be stolen or misused.

2. Prevention of Financial Loss :-

Cyber attacks like hacking and phishing can lead to loss of money.

Cyber security helps prevent:

Online fraud

Unauthorized transactions

Business losses

3. Protection Against Cyber Attacks :-

Cyber security systems protect against

Malware

Hacking

Ransomware

Phishing attack.

6. Methods to Prevent Cyber Attacks :

Preventing cyber attacks is very important to protect computers, networks, and personal data. By following proper security methods, we can reduce the risk of hacking, data theft, and other cyber threats.

1. Use Strong Passwords :-

Use a combination of letters, numbers, and symbols

Avoid simple passwords like “123456”

Change passwords regularly

2. Enable Two-Factor Authentication (2FA) :-

Adds an extra layer of security

Requires OTP or biometric verification

Even if the password is stolen, the account stays protected

3. Install Antivirus Software :-

Detects and removes malware

Protects system from viruses and harmful files

4. Use Firewalls :-

Blocks unauthorized access to your network

Acts as a barrier between trusted and untrusted networks

5. Keep Software Updated :-

Install updates regularly

Fixes security weaknesses (bugs)

6. Be Aware of Phishing Attacks :-

Do not click on unknown links

Avoid sharing personal information on suspicious websites.

7. Advantages of Cyber Security :

Cyber security is the practice of protecting systems, networks, and programs from digital attacks, data breaches, and unauthorized access. In today's digital world, cyber security is critical for individuals, businesses, and organizations. Implementing strong cyber security measures offers numerous advantages, which are detailed below:

1. Protection of Sensitive Data :-

Cyber security safeguards confidential information, such as personal data, financial records, medical records, and intellectual property.

Prevents unauthorized access, identity theft, and data leaks.

Ensures that sensitive data remains private and secure, which is especially crucial for businesses and government organizations.

2. Prevention of Cyber Attacks :-

Strong cyber security measures reduce the risk of cyber attacks, such as malware, phishing, ransomware, and hacking.

Protects systems from disruption, financial loss, or reputational damage.

Organizations can continue smooth operations without interruptions caused by cyber threats.

3. Safeguards Financial Transactions :-

In the digital age, many transactions are online (banking, e-commerce, payment gateways).

Cyber security protects against fraudulent transaction, online scams, and financial then left

Ensure safe and secure online payments, which increases trust in digital services

8. Implementation of Cyber Security :

The implementation phase is where the planned system design and methodology are converted into a working cyber security solution. This phase includes installation, configuration, coding, testing, and deployment.

1.Environment Setup :-

Hardware Requirements:

Computer / Laptop with minimum 8 GB RAM, 500 GB storage.

Internet connection for cloud tools and updates.

Software Requirements:

Operating System: Windows 10 / Linux.

Programming Languages: Python / Java / C#.

Database: MySQL / MongoDB for storing user and system data.

Security Tools: Wireshark, Nmap, OpenVAS, or custom scripts.

UI Prototype Tool: Figma for designing dashboards and screens.

2. Module-wise Implementation :-

a) Login / Authentication Module

Users create an account with username/email and password.

Passwords are encrypted using SHA-256 or bcrypt.

Two-factor authentication (2FA) is implemented for additional security.

Implementation Steps:

4. Maintains Business Reputation :-

Data breaches or cyber attacks can harm an organization's credibility.

Cyber security ensures customer confidence and trust, which is essential for long-term business success.

Organizations with strong cyber security can demonstrate reliability and professionalism to clients, partners, and stakeholders.

5. Compliance with Legal Requirements :-

Many countries have data protection laws and regulations, such as GDPR, HIPAA, or IT Act.

Cyber security ensures compliance, avoiding legal penalties, fines, or lawsuits.

Helps businesses meet industry standards for data protection and security.

6. Protection Against Data Loss :-

Cyber security measures, including backup systems and disaster recovery, prevent permanent data loss.

In case of accidental deletion, ransomware, or hardware failure, data can be restored safely.Ensures business continuity even during cyber incidents.

7. Enhanced Productivity :-

A secure digital environment allows employees to work without interruptions caused by cyber threats.Systems remain stable, reducing downtime and improving efficiency.Teams can focus on core business activities instead of dealing with cyber attacks.Promotes Safe Use of Technology

Cyber security educates individuals and organizations about safe online practices, like strong passwords, avoiding phishing scams, and secure browsing.

Protects users from malware infections, spyware, and identity theft.Encourages responsible and secure use of digital platforms.

Supports Emerging Technologies

With new technologies like cloud computing, AI, IoT, and blockchain, cyber threats are evolving.Cyber security ensures safe adoption of these advanced technologies.

Enables organizations to innovate and scale digitally without compromising security.

8. Reduces Financial Loss :-

Cyber attacks can result in huge financial losses due to theft, ransom payments, or recovery costs.

Effective cyber security reduces the risk and cost of potential cyber threats.

Organizations can save money, resources, and reputation by investing in proactive cyber security measuresThe advantages of cyber security include data protection, prevention of cyber attacks, safe financial transactions, legal compliance, enhanced productivity, and long-term business reputation. In a world increasingly dependent on digital systems, strong cyber security is essential for individuals, businesses, and governments to maintain trust, privacy, and operational stability.

9.Output of Cyber Security :

The output section demonstrates the working results of the implemented cyber security system. It shows how the system protects data, monitors threats, and manages users effectively. The output can be both visual (screens/dashboards) and data-driven (logs, alerts, reports).

1.Login / Authentication Output :-

Successful Login:

When valid credentials are entered, the user is redirected to the dashboard.

Two-factor authentication prompt appears if enabled.

Error Handling:

Displays error messages for incorrect username/password or unauthorized access attempts.

Output Example:

"Login Successful" message with dashboard redirect.

Alerts for failed login attempts or suspicious activity.

2.Dashboard / Home Page Output :-

System Overview:

Displays real-time security alerts, total users, active sessions, and system health status.

Visual Output:

Charts showing attempted attacks, login trends, and threat statistics.

Color -coded notification for high, medium, and low severity aleets.

User interaction:

Clickable widgets to navigate to thrate details, reports, or use management.

3.Threat Detection Output :-

Detected Threats:

Logs of suspicious activity such as malware detection, unauthorized login attempts, or network anomalies.

Visual Output:

Table or dashboard list of detected threats with:

Threat type

Source IP or user

Severity level

Timestamp

Notification Output:

Pop-up alerts or messages for high-severity threats.

Example:

"Suspicious login attempt detected from IP 192.168.1.100 – Blocked successfully."

4.Data Protection Output :-

Secure Data Storage:

Passwords and sensitive data stored in encrypted form.

Backup Status:

Successful automated backups displayed with date and time.

Audit Output:

Logs showing who accessed sensitive data and when.

5. Use Management output:

Admin panel:

Lists all user, their roles, and status (active/inactive)

Button for add, remove, or modify users

Example:

"User JohnDoe added successfully with role 'Author'."

Access Logs:

Shows login history, failed attempts, and user activity.

6.Reports / Analytics Output :-

Reports Generated:

Weekly or monthly reports for security events, login attempts, and system activity.

Graphical Output:

Pie charts for threat distribution.

Line charts for login trends.

Tables for detailed event logs.

Export Options:

Reports can be exported in PDF, Excel, or CSV format for auditing.

7.Cyber Security Index Output :-

Scorecard Display:

Security score based on metrics like network security, data protection, user awareness, and threat response.

Example Output:

Network security: 90/100

Data protection: 95/100

Threat detection: 95/100

Overall cyber security index: 90/100

Visual output:

bar graphs or radar charts to compare different security aspect.

10.Conclusion :

The Cyber Security Project successfully demonstrates the importance, implementation, and impact of cyber security in today's digital world. With the increasing reliance on digital systems, data storage, and online transactions, protecting information from unauthorized access, cyber attacks, and data breaches has become essential for both individuals and organizations.

Through this project, we have implemented a robust system that covers multiple aspects of cyber security, including login authentication, threat detection, data protection, user management, reporting, and analytics. Each module is designed to prevent, detect, and respond to cyber threats effectively, ensuring the confidentiality, integrity, and availability of data.

The project also demonstrates the creation of a Cyber Security Index, which provides a quantitative measure of the system's security performance, helping in continuous monitoring and improvement. By generating real-time alerts, visual dashboards, and reports, the system enables administrators and users to take proactive actions against potential threats.

In addition to the current functionalities, this project highlights the future potential of cyber security systems, including AI-based threat detection, cloud security, IoT protection, and blockchain-enabled authentication. These enhancements can make the system even more intelligent, adaptive, and resilient against evolving cyber threats.

Overall, this project proves that effective cyber security is not just a technological requirement but a critical necessity in today's digital age. It ensures the safe usage of digital platforms, protects sensitive information, and builds trust among users, making it a valuable contribution to modern IT infrastructure.

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